Aman Joshi

AI/ML ENGINEER

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Github:AMANj2002 Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer experienced in **machine learning, deep learning, Generative AI, and LLM application development**. Skilled in Python, PyTorch, model fine-tuning, RAG pipelines, vector databases, LLM inference optimization, and computer vision, with hands-on experience building scalable AI solutions, deploying models, and developing production-ready AI APIs using FastAPI and Docker.

PROFESSIONAL EXPERIENCE

eDOT Solutions, *AI/ML Engineer* 04/2025 – Present

- Built production-grade **RAG and Contextual RAG pipelines** using vector databases, hybrid retrieval, and prompt engineering.
- Deployed scalable LLM inference infrastructure using **vLLM, LiteLLM, FastAPI, Docker, and GPUs**.
- Fine-tuned open-source models using **Unsloth, LoRA, QLoRA, and dLLM**, including Qwen, LLaDA, and Gemma.
- Developed **OCR, VLM, YOLO, and real-time video analytics** solutions using NVIDIA Jetson, DeepStream, and GStreamer.
- Optimized model inference through **quantization, GPU optimization, and edge deployment**.
- Automated data annotation using **Label Studio** and integrated **Langfuse** for LLM observability and evaluation.
- Built and deployed scalable **AI APIs and backend services** using Python, FastAPI, and Docker.

Persistent Systems Ltd., *Student Intern* 07/2024 – 09/2024

- Developed and maintained **responsive web applications** using **Node.js, React.js, Next.js, and Tailwind CSS**, including an online booking platform.
- Gained hands-on experience in **professional software development**, collaborating with team members and contributing to project goals.

EDUCATION

BE(Information technology), *PCCE* 2021 – 2025

- **Academic Achievement:** Secured an overall **CGPA of 8.7**.
- **Practical Experience:** Actively participated in multiple **AI and Web development hackathons**.

HSSC, *Deepvihar higher secondary school* 2019 – 2021

Percentage:66%
Won multiple awards in **project-making competitions**
Active member of **science club**

SSC, *Vidya mandir secondary school* 2009 – 2019

percentage:70%
Leader of **chess team** for **2 years**
Active member of **science club**

SKILLS

Programming Languages: (Python, SQL, C++, JavaScript, TypeScript), **LLM Serving & MLOps** (vLLM, LiteLLM Proxy, LangFuse, FastAPI, Docker, REST APIs, Linux, GPU Optimization, Model Quantization), **AI / Machine Learning** (PyTorch, Hugging Face Transformers, LangChain, RAG, Contextual RAG, Prompt Engineering, LoRA, QLoRA, Unsloth, dLLM, Model Fine-tuning, LLM Pretraining), **Computer Vision** (YOLO, OCR, Vision-Language Models (VLMs), Object Detection, NVIDIA Jetson, DeepStream, GStreamer), **Databases** (FAISS, Pinecone, PostgreSQL, MongoDB, MySQL,Qdrant), **Agentic Systems** (Agentic AI, Multi-Agent Systems, Autonomous AI Agents, Tool Use / Function Calling), **Data Science** (Pandas, NumPy, scikit-learn), **Cloud / Deployment** (AWS, GCP, Azure), **CI/CD & DevOps** (Jenkins, GitHub Actions, Azure DevOps)

PROJECTS

AI-Powered Courtroom Simulation, *Final Year Project*

- Developed an **LLM-powered virtual courtroom** using **Gemini** and a fine-tuned **Mistral 7B** model to detect misinformation on social media.
- Designed a **multi-agent system** with lawyer and judge agents, leveraging real-time web search and **RAG** with **FAISS/Pinecone** for evidence retrieval.
- Built the backend using **Flask** and the frontend using **Expo**, delivering an interactive AI-powered application.
- **Research work based on this project was published in an IEEE conference/publication.**

Medilens (Flask,Flutter) , *Hackathon*

- Developed **Medilens**, a **Flutter** and **Flask**-based mobile app for **medicine identification and information retrieval**.
- Users can **photograph medicines** to get details on **side effects, uses, and other relevant information**.
- **Fine-tuned Gemini** was used to fetch accurate **side effects, uses, and other relevant medicine details**.
- Showcased expertise in **cross-platform mobile development** and **backend integration** for a seamless user experience.

DataDashBot for Ecommerce & MediMentor for Healthcare, *Won 2nd place*

- Developed AI chatbots: **DataDashBot** for E-commerce and **MediMentor** for Healthcare during a hackathon.
- Played a key role in **design and implementation**, enhancing **customer engagement** and **healthcare delivery**.
- Utilized a **fine-tuned GPT model** and directly integrated its **API** for chatbot functionalities.
- Secured **2nd place**, demonstrating **innovation, technical excellence, and problem-solving skills**.

Fake News Detection In Social Media(Flutter,Flask) , *24 Hours hackathon*

- Secured **2nd place at GEC Hackathon** with the "Fab-Four" team.
- Developed an Android Fact-Checking App using **Flutter**, **Flask**, and **machine learning**.
- Recognized for innovation and effectiveness under pressure.

Laya Gateway , *Open-Source Project*

- Developed a lightweight LLM gateway using **Python, Laya, and Hugging Face** for prompt guardrails and intelligent model routing.
- Implemented confidence-based safety classification (**allow, review, block**) and model selection using **Pydantic** and LLM inference.
- Optimized low-latency, CPU-based inference for efficient prompt classification and routing using **non-autoregressive models**.

PUBLICATIONS

Application for Identifying False Information in Social Media Chats

2025 International Conference on Modeling, Simulation & Intelligent Computing (MoSICom), IEEE Xplore
Published: March 2026 | **DOI:** 10.1109/MoSICom67259.2025.11398314